Ising Model: Theory and Monte Carlo Simulation

Botond Koroknai

Abstract

This study evaluates the performance and numerical stability of the Metropolis-Hastings algorithm in simulating the Ising model across 1D and 2D lattice geometries. We implement a Monte Carlo (MC) framework utilizing importance sampling to explore the state space, employing local energy updates to reduce the per-flip computational complexity to $\mathcal{O}(1)$. For the 1D system, the simulation's convergence is validated against the exact Transfer Matrix solution. In the 2D case, we utilize Periodic Boundary Conditions (PBC) to approximate bulk thermodynamic properties on a finite $N \times N$ grid. The numerical analysis focuses on the emergence of criticality near the Onsager temperature (T_c), where we quantify the divergence of second-order moments—specifically the specific heat and magnetic susceptibility. We further examine the limitations of the Metropolis update in small-scale lattices ($N = 20$), identifying non-physical artifacts and "false peaks" caused by global lattice flipping and finite-size scaling effects. Our results demonstrate that while the algorithm robustly identifies the phase transition, the precision of critical exponents is constrained by the lattice size and the statistical variance inherent in importance sampling.

1 The Ising Model

The model was introduced by Ernst Ising and Wilhelm Lenz, with the motivation to explain ferromagnetism. In this discrete model, each spin s_i can exist in one of only two states:

$$s_i \in \{+1, -1\} \tag{1}$$

These states represent a magnetic moment pointing up $(+1)$ or down (-1).

Ferromagnetism

Ferromagnetism occurs when internal interactions cause spins to align spontaneously in the same direction and this behaviour is governed by the coupling constant. When the system undergoes such a transitions from a disordered arrangement to an ordered one, it moves from a paramagnetic state to a ferromagnetic state, creating a macroscopic magnetic moment.

2 The Hamiltonian

The total energy of such a system can be described by its Hamiltonian (H), which accounts for interactions between neighbors and the influence of an external environment:

$$H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i \tag{2}$$

Where the variables are defined as follows:

- J (coupling constant): This parameter controls the strength and nature of the interaction between neighboring spins.
  - $J > 0$: Ferromagnetic interaction - neighbors align to minimize energy
  - $J < 0$: Antiferromagnetic interaction- neighbors point in opposite directions.
- h (external magnetic field): This represents an external force that biases the spins. A positive h encourages spins to point up $(+1)$, while a negative h biases them toward down (-1).
- $\langle i, j \rangle$**:** To reduce the computational cost, the summation is performed only over nearest-neighbor pairs only.

3 The 1D Ising Model

In a 1D model, each spin (s_i) only interacts with its two neighbors. This geometry simplifies the Hamiltonian sum to spin-pairs $(i, i+1)$, providing a mathematically tractable environment to study the fundamental principles of statistical mechanics before moving to higher dimensions.

4 Physical Mechanisms

The macroscopic state of the Ising model is dictated by a fundamental struggle between two opposing forces: internal coupling and thermal noise. The balance between these forces determines whether the system exists in an ordered or disordered state.

4.1 Coupling Energy and the Drive for Order

The interaction term of $-Js_is_j$ represents the system's drive toward minimizing internal energy. In a ferromagnetic system ($J > 0$), energy is minimized when neighboring spins are parallel ($s_i = s_j$). This force acts as a coordinating mechanism, attempting to organize the lattice into a coherent ordered (ferromagnetic) phase, characterized by the formation of large macroscopic magnetic domains.

4.2 Thermal Fluctuations and Entropy

Opposing the internal coupling is the thermal energy, represented by k_BT. This force introduces entropy into the system. As the temperature increases, thermal fluctuations provide the energy necessary for spins to overcome the exchange interaction and flip randomly. This process disrupts the alignment, leading to a disordered (paramagnetic) phase where the net magnetization is zero and energy fluctuations are frequent.

4.3 Helmholtz Free Energy

The state of the system at any given temperature is determined by the minimization of the Helmholtz Free Energy (F), defined as:

$$F = E - TS \tag{3}$$

where E is the internal energy, T is the temperature, and S is the entropy. Interpreting this formula, it essentially provides the mathematical criteria for the phase behavior of the system:

- At Low Temperatures: The TS term is negligible. The system prioritizes the minimization of internal energy (E), leading to a stable, aligned configuration.
- At High Temperatures: The TS term becomes the dominant factor. To minimize F, the system must maximize entropy (S), causing the ordered structure to dissolve into a state of maximum randomness.

5 Thermodynamic Observables

To characterize each macrostate of the system, four key quantities will be observed as functions of temperature (T). Examining these observables we could gather quantitative evidence for the transition between the ordered and disordered phases.

5.1 Internal Energy and Magnetization

The primary indicators of the system's state are its average energy density and its degree of net alignment:

1. Average energy per spin ($\langle e \rangle$): This represents the internal energy density of the simulated lattice. At $T = 0$, a ferromagnetic system ($J > 0$) occupies its ground state where all spins are parallel, resulting in a minimum energy of $-z \cdot \frac{J}{2}$ per spin (where z is the number of neighbors). As thermal energy increases, stochastic spin flips against the local alignment raise the total energy toward zero.

$$\langle e \rangle = \frac{\langle H \rangle}{N} \tag{4}$$

2. Magnetization per Spin ($\langle |m| \rangle$): Magnetization serves as the *order parameter* for the Ising model. In finite-size simulations, we typically measure the *absolute* magnetization to prevent the global flipping of the lattice from averaging the result to zero:

$$\langle |m| \rangle = \frac{1}{N} \left\langle \left| \sum_i s_i \right| \right\rangle \tag{5}$$

A value of $\langle |m| \rangle \approx 1$ signifies a highly ordered state, whereas $\langle |m| \rangle \approx 0$ represents a paramagnetic state.

5.2 Response Functions and the Fluctuation-Dissipation Theorem

Specific Heat and Magnetic Susceptibility are *response functions* that describe how the system's equilibrium state reacts to external perturbations.

1. The Specific Heat (C_v) measures the sensitivity of the internal energy to temperature changes and is proportional to the energy variance:

$$C_v = \frac{\partial \langle e \rangle}{\partial T} = \frac{1}{k_B T^2} \left(\langle E^2 \rangle - \langle E \rangle^2 \right) \tag{6}$$

In the 1D model, C_v exhibits a smooth *Schottky anomaly.* In the 2D model, this function becomes the primary indicator of the phase transition, developing a sharp peak that diverges at the critical temperature in the thermodynamic limit.

2. The magnetic Susceptibility (χ) quantifies the magnetization's response to a small external field (h). In the absence of an actual field in the simulation, we calculate it from magnetization fluctuations:

$$\chi = \frac{\partial \langle m \rangle}{\partial h} \Big|_{h=0} = \frac{1}{k_B T} \left(\langle M^2 \rangle - \langle M \rangle^2 \right) \tag{7}$$

At the critical point, the susceptibility diverges, signifying that the system is extremely sensitive to external perturbations.

5.3 Numerical Constraints

In numerical simulations where N is finite (e.g., $N = 20$), several artifacts must be accounted for:

- Spontaneous symmetry breaking: In small systems at low T, the energy barrier between the "all-up" and "all-down" states is finite. This could cause that the lattice flip its global direction during a measurement, leading to a dip in the absolute magnetization averages.
- Peak rounding and shifting: The sharp divergences expected at a phase transition are often "rounded" or slightly shifted in temperature when working with finite lattices.

6 The Monte Carlo Method

A system with N spins has 2^N possible configurations, therefore a modest lattice of $N = 100$, this results in $2^{100} \approx 10^{30}$ states, making the summation computationally impossible.

6.1 The Boltzmann Distribution

In thermal equilibrium, the probability of the system being in a specific state with energy H is given by the Boltzmann distribution:

$$P(\text{config}) = \frac{e^{-H/k_B T}}{Z} \tag{8}$$

where Z is the partition function (the sum over all possible states) and k_B is the Boltzmann constant. This distribution is the mathematical realization of the competition between energy and entropy:

- In low energy states the exponential factor of $e^{-H/k_B T}$ is large (aligned spins), making them highly probable.
- Thermal weighting: The denominator T in the exponent determines how strictly the system adheres to the low-energy rule. At high T, the exponent approaches zero, meaning the term $e^0 = 1$ becomes similar for all states, leading to the uniform probability characteristic of a disordered system.

By using the Metropolis algorithm, we generate a sequence of configurations that naturally follow this distribution without ever needing to calculate the impossible sum Z.

6.2 The Metropolis-Hastings Algorithm

The simulation proceeds through the iterative steps defined by Metropolis et al. [2]:

1. It selects a random spin s_i.
2. Calculates the energy change ΔE that would result from flipping that spin.
3. Accepts the flip with a probability P_{accept} defined by:

$$P_{\text{accept}} = \min\left(1,\ e^{-\Delta E/k_B T}\right) \tag{9}$$

4. Repeats this process N times to constitutes one **sweep**.
5. Then it performs many sweeps to reach equilibrium and calculate thermodynamic averages.

7 Analytical solution

For $h = 0$, this leads the exact solutions used to validate our Monte Carlo results:

1. Internal energy per spin:

$$\langle e \rangle = -J \tanh\left(\frac{J}{k_B T}\right) \tag{10}$$

2. Specific heat:

$$C_v = k_B \left(\frac{J}{k_B T}\right)^2 \operatorname{sech}^2\left(\frac{J}{k_B T}\right) \tag{11}$$

   The absence of a divergence in C_v confirms the lack of a phase transition for $T > 0$.

3. Correlation length (ξ):

$$\xi = \frac{-1}{\ln\left[\tanh\left(\frac{J}{k_B T}\right)\right]} \tag{12}$$

8 Results and Discussion

The simulation was performed at a temperature of $T = 2.0$ (in units where $k_B = 1, J = 1$). At this temperature, the thermal energy is twice the strength of the coupling constant, placing the system in the paramagnetic (disordered) regime. The numerical results obtained are as follows:

Observable	**Simulation Value**	**Theoretical Expectation**
Energy per spin $\langle e \rangle$	-0.4476	≈ -0.4621
Magnetization $\langle \lvert m \rvert \rangle$	0.1156	0.0 (Infinite limit)
Specific Heat C_v	0.2151	≈ 0.1966
Susceptibility χ	1.0996	> 0

8.1 Energy and Magnetization

The simulated energy per spin ($\langle e \rangle = -0.4476$) aligns closely with the exact analytical value ($e_{exact} \approx -0.4621$), confirming successful equilibration via the Metropolis algorithm. The minor 3% discrepancy is a standard consequence of finite-size effects ($N = 100$) and limited sampling statistics. Measured magnetization ($\langle \lvert m \rvert \rangle = 0.1156$) further confirms the expected disordered state, with small non-zero values arising from natural thermal fluctuations in a finite chain.

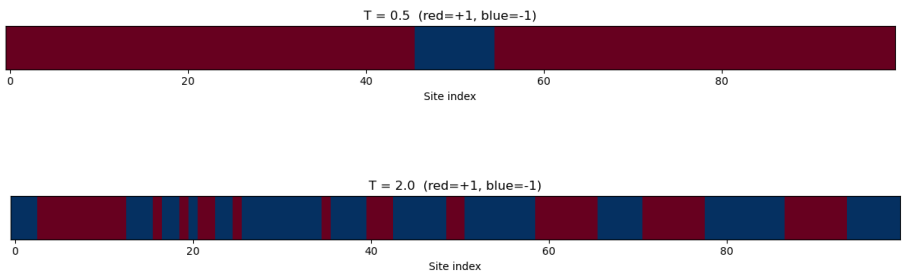
8.2 Response functions and fluctuations

Specific heat ($C_v = 0.2151$) and susceptibility ($\chi = 1.0996$) were derived via the fluctuation-dissipation theorem.

- Specific heat: The value aligns with the theoretical 0.1966, placing the system on the descending slope of the *Schottky anomaly*. As T increases, C_v decreases as the system approaches maximum entropy.
- Susceptibility: The non-zero χ confirms that despite being in a disordered phase, spins remain coupled ($J = 1.0$) and maintain short-range collective responsiveness to their neighbors.

9 Visual analysis

To complement the thermodynamic averages, we visualize the actual state of the spin chain at varying temperatures. This allows for a qualitative assessment of correlation lengths and magnetic domains.



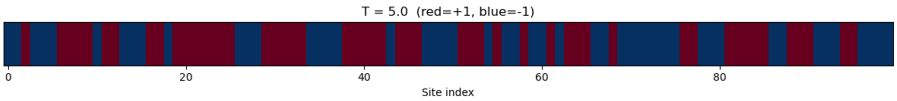


Figure 1: Visual representation of the 1D Ising chain at $T = 0.5$, $T = 2.0$, and $T = 5.0$. Red blocks represent $s_i = +1$ and blue blocks represent $s_i = -1$.

9.1 Low Temperature ($T = 0.5$)

At $T = 0.5$, the thermal energy ($k_B T$) is significantly lower than the exchange energy J. Consequently, the system is dominated by the coupling term of the Hamiltonian.

- Observations: The visualization shows large, contiguous blocks of color (either all red or all blue).
- Physics: These are known as *magnetic domains*. The correlation length ξ is large, meaning a spin's orientation strongly influences its neighbors many sites away.

9.2 Intermediate Temperature ($T = 2.0$)

At $T = 2.0$, the energy and entropy are in direct competition.

- Observations: The large domains begin to fracture. We see a mixture of medium-sized clusters and individual spin flips (noise).
- Physics: This is the regime of our previous numerical analysis ($\langle e \rangle \approx -0.44$). The system is paramagnetic, but short-range correlations still persist. The "islands" of alignment are temporary and constantly shifting.

9.3 High Temperature ($T = 5.0$)

At $T = 5.0$, thermal fluctuations completely overwhelm the coupling constant J.

- Observations: the pattern in the chain wildly mixes.
- Physics: The correlation length ξ approaches zero. Each spin acts almost independently of its neighbor, leading to a state of maximum entropy. The net magnetization $\langle m \rangle$ fluctuates very closely around zero, and the internal energy $\langle e \rangle$ approaches its theoretical maximum of 0.

10 Equilibration and Time Series Analysis

In Monte Carlo simulations, it is essential to verify that the system has reached *thermal equilibrium* before calculating the thermodynamic averages. By plotting the time series of the energy and magnetization, we can observe the stochastic evolution of the system over successive Monte Carlo sweeps.

10.1 Thermal Equilibrium and Fluctuations

The time series at $T = 2.0$ demonstrates two critical features of the Metropolis algorithm in the steady state:

- The energy and magnetization fluctuate around stable mean values ($\langle e \rangle \approx -0.45$ and $\langle m \rangle \approx 0$). The lack of an upward or downward trend indicates that the equilibration phase was successful, and the system is now sampling from the true Boltzmann distribution.

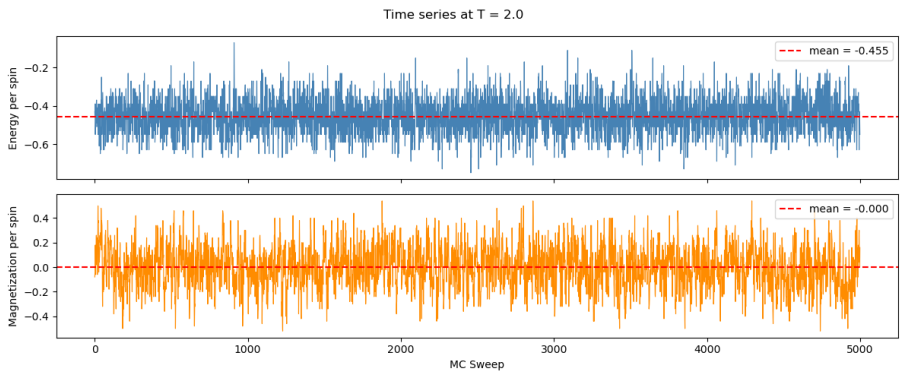
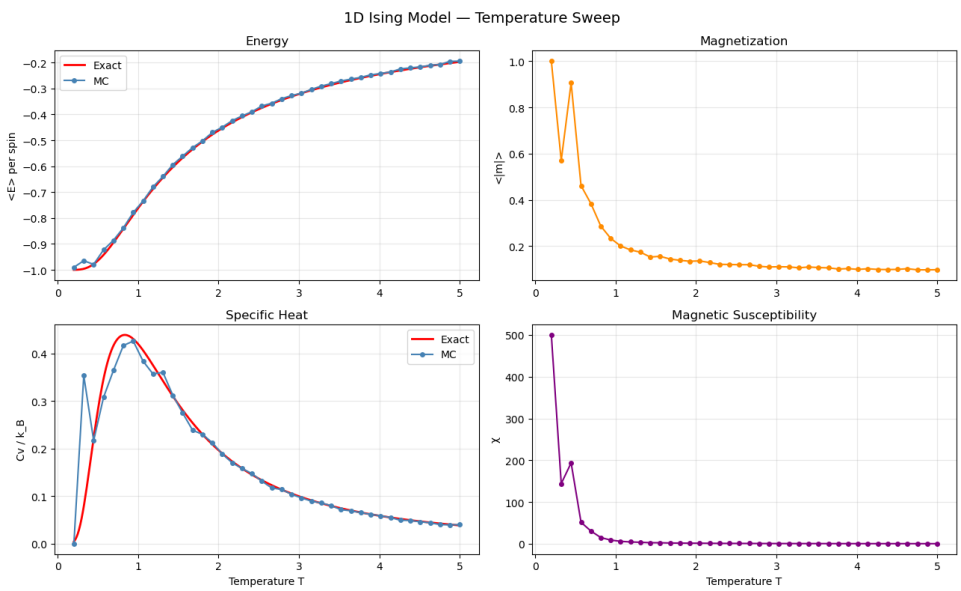


Figure 2: Time series of energy and magnetization per spin at $T = 2.0$ for $N = 100$.

- Noise: The "width" of the fluctuations in the energy plot is directly related to the specific heat (C_v), while the fluctuations in magnetization are related to the magnetic susceptibility (χ). At $T = 2.0$, these fluctuations are significant, indicating that the thermal energy $k_B T$ is high enough to frequently flip spins against the coupling J.

11 Temperature sweep and comparative analysis

To validate the Monte Carlo simulation, a temperature sweep was performed across the range of $T \in [0.2, 5.0]$. This range covers the transition from a highly correlated state to a nearly random state. For each temperature point, the system was equilibrated for 1,000 sweeps followed by 3,000 measurement sweeps to ensure statistical reliability.



11.1 Comparison with exact solutions

The simulation results for Energy (E) and Specific Heat (C_v) are plotted against the exact solutions.

- Energy (E): The Monte Carlo data points (blue) show an excellent fit with the exact analytical curve (red). At $T \to 0$, the energy approaches $-J = -1$, representing the

perfectly aligned ground state. As temperature increases, the energy rises monotonically toward zero.

- Specific heat (C_v): The simulation accurately captures the *Schottky anomaly*, the broad peak is centered around $T \approx 0.8$ and it represents the temperature at which the system undergoes the greatest internal structural change as bonds begin to break more frequently.

11.2 Magnetic behavior and susceptibility

While E and C_v have exact 1D solutions for comparison, the magnetization (M) and susceptibility (χ) provide insight into the phase behavior of the finite $N = 100$ lattice.

- Magnetization ($\langle |m| \rangle$): It is important to note that in an infinite 1D Ising model, magnetization is zero for all $T > 0$. The non-zero values observed at $T < 1.0$ are a result of finite-size effects. In a chain of only 100 spins, the system can remain in a single domain for long periods, creating an "apparent" order that would vanish in a larger system.

- Magnetic Susceptibility (χ): The susceptibility shows a massive increase as $T \to 0$. This indicates that at low temperatures, the spins are so highly correlated that even a infinitesimal external field could cause the entire chain to flip. This "divergence" at $T = 0$ is consistent with the fact that the 1D Ising model only possesses a phase transition at absolute zero.

12 The 2D Ising Model

Having verified our Monte Carlo framework against the 1D analytical solution, we extend the simulation to a two-dimensional square lattice. In this configuration, the geometry shifts from a linear chain to an $N \times N$ grid.

12.1 Why 2D is different?

The fundamental difference between 1D and 2D lies in the energy cost of disorder. In a 1D chain, a single spin flip can create a "break" (domain wall) that separates the system into two independent sections with an energy cost of only $2J$. Because this cost is constant regardless of system size, thermal entropy always dominates, preventing long-range order.

In a 2D lattice, creating a region of opposite spins (a domain) requires breaking bonds along the entire perimeter of that region. The energy cost of disorder now scales with the size of the domain. This creates a significant energy barrier that thermal fluctuations cannot easily overcome at low temperatures, allowing a stable, ordered *ferromagnetic phase* to exist until the critical temperature T_c is reached.

12.2 The Onsager solution and spontaneous symmetry breaking

Unlike the 1D model, which remains disordered at all finite temperatures, the 2D Ising model exhibits a second-order phase transition at a non-zero critical temperature (T_c). This was first proven analytically by Lars Onsager in 1944. For a system with $J = 1$ and $k_B = 1$, the exact transition point is:

$$T_c = \frac{2J}{k_B \ln(1 + \sqrt{2})} \approx 2.269 \tag{13}$$

At this critical point, the system undergoes spontaneous symmetry breaking. Below T_c, the internal coupling is sufficient to overcome thermal noise, allowing the spins to choose a preferred orientation (all "Up" or all "Down") even in the absence of an external field. Above T_c, this

order vanishes into a paramagnetic state. In our 20×20 simulation, we expect to observe the marks of this transition.

12.3 Physical regimes and phase behavior

The system's thermodynamic behavior is divided into two distinct regimes by the critical temperature T_c:

- Ferromagnetic phase ($T < T_c$): The exchange interaction J dominates thermal noise, leading to spontaneous symmetry breaking. In this regime, spins align into macroscopic domains, resulting in a non-zero magnetization $\langle |m| \rangle$ without the need for an external field.
- Paramagnetic phase ($T > T_c$): Thermal energy overwhelms the coupling forces, causing the correlated domains to dissolve. The system enters a state of maximum entropy where the net magnetization is zero and spins fluctuate independently.

13 Computational implementation

The transition from a 1D chain to a 2D square lattice requires a shift in how we handle spatial indexing and boundary conditions. We represent the system as an $N \times N$ matrix where each element corresponds to a spin $s_{i,j}$.

13.1 Periodic boundary conditions

To approximate an infinite system with a finite $N \times N$ lattice, we implement periodic boundary conditions. This ensures that every spin maintains a constant coordination number of 4 by linking opposing edges.

13.2 The 2D Monte Carlo sweep

In the 2D model, the definition of a "sweep" changes to reflect the increased number of degrees of freedom. Now a single Monte Carlo sweep is defined as N^2 attempted flips. This ensures that, on average, every spin in the $N \times N$ lattice has had the opportunity to flip once.

The algorithm follows these programmatic steps:

1. Creates a random $N \times N$ matrix of spins ($+1$ and -1) is generated.
2. Chooses a random coordinate (i, j).
3. Calculates the energy change ΔE. If $\Delta E \leq 0$, the flip is accepted. If $\Delta E > 0$, the flip is accepted with a Boltzmann probability $e^{-\Delta E/T}$.
4. This is repeated N^2 times to complete one sweep.

14 Visual Analysis

To understand the microscopic behavior of the 2D system, we visualize the 10×10 lattice configurations at three key temperatures: $T = 1.0$ (Ferromagnetic phase), $T = 2.269$ (Critical point), and $T = 4.0$ (Paramagnetic phase).

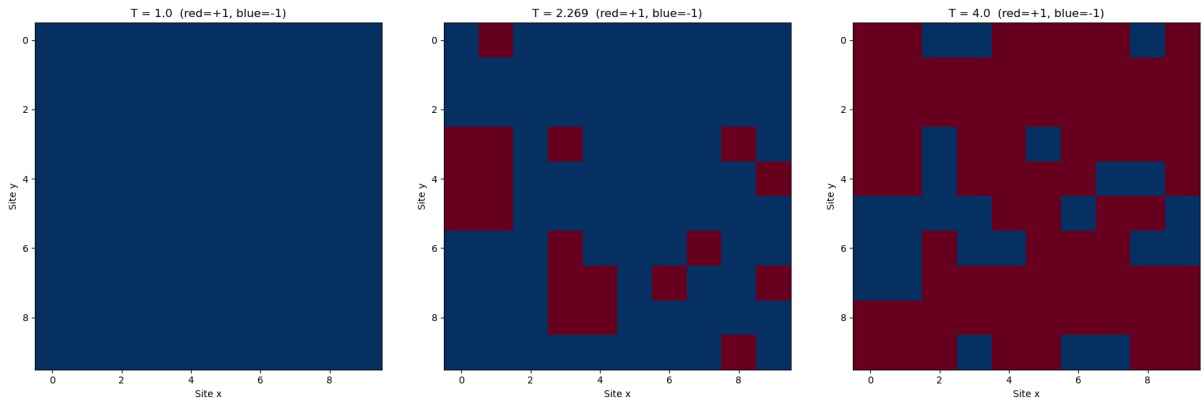


Figure 3: *
$T = 1.0$

Figure 4: *
$T = 2.269$ (T_c)

Figure 5: *
$T = 4.0$

Figure 6: Snapshots of the 2D Ising lattice showing the transition from large ordered domains to a disordered paramagnetic state. Red pixels represent $s_i = +1$ and blue pixels represent $s_i = -1$.

14.1 Phase Descriptions

- Ordered phase ($T = 1.0 < T_c$): At low temperatures, the system is dominated by huge, uniform clusters. The lattice "chooses" a majority orientation (either mostly red or mostly blue), resulting in a high absolute magnetization.
- Critical Point ($T = 2.269 \approx T_c$): At the critical temperature, we observe fractal-like clusters. Domains of all sizes—from single spins to system-spanning structures—coexist. This scale-invariance is the hallmark of criticality. The correlation length ξ effectively reaches the size of our lattice, making the system extremely sensitive to fluctuations.
- Disordered Phase ($T = 4.0 > T_c$): As the temperature increases, thermal noise shreds the domains into a well mixed pattern where each spin behaves nearly independently, and the net magnetization is zero.

15 2D Dynamics at the Critical Point

Visualizing the time series of the 2D Ising model at the critical temperature ($T_c \approx 2.269$) provides a window into the complex dynamics of phase transitions. Unlike the stable fluctuations seen at high or low temperatures, the system at T_c exhibits unique stochastic behavior.

15.1 Critical Fluctuations

At T_c, the energy and magnetization do not settle into a narrow band of noise. Instead, we observe large-amplitude fluctuations.

- Energy Fluctuations: The variance in the energy trace is at its maximum here, which corresponds to the peak in the Specific heat (C_v).
- The magnetization trace often exhibits metastable behavior, where the system stays near a certain value before swinging wildly to another. This is a result of the magnetic susceptibility (χ) approaching a maximum. In this state a random flip can trigger a change in the global magnetization.

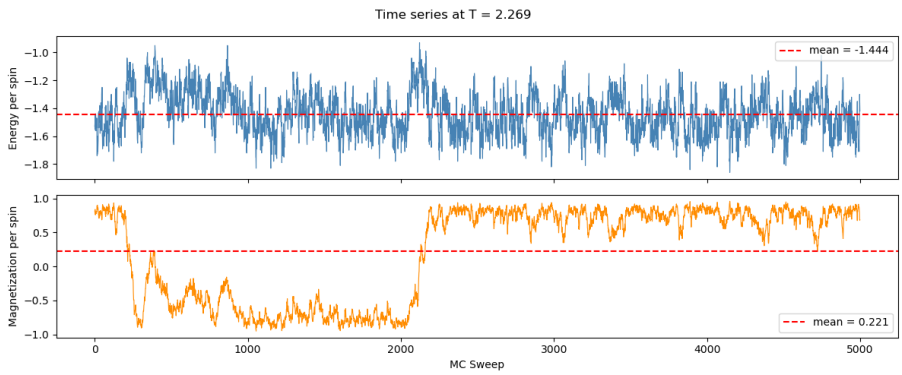


Figure 7: Time series of energy and magnetization for a 20×20 lattice at $T = 2.269$.

16 2D Temperature sweep

The 2D Ising model was simulated across 40 temperature points ranging from $T = 0.5$ to $T = 4.0$. This sweep allows for a direct observation of the phase transition and a comparison with the exact Onsager solution.

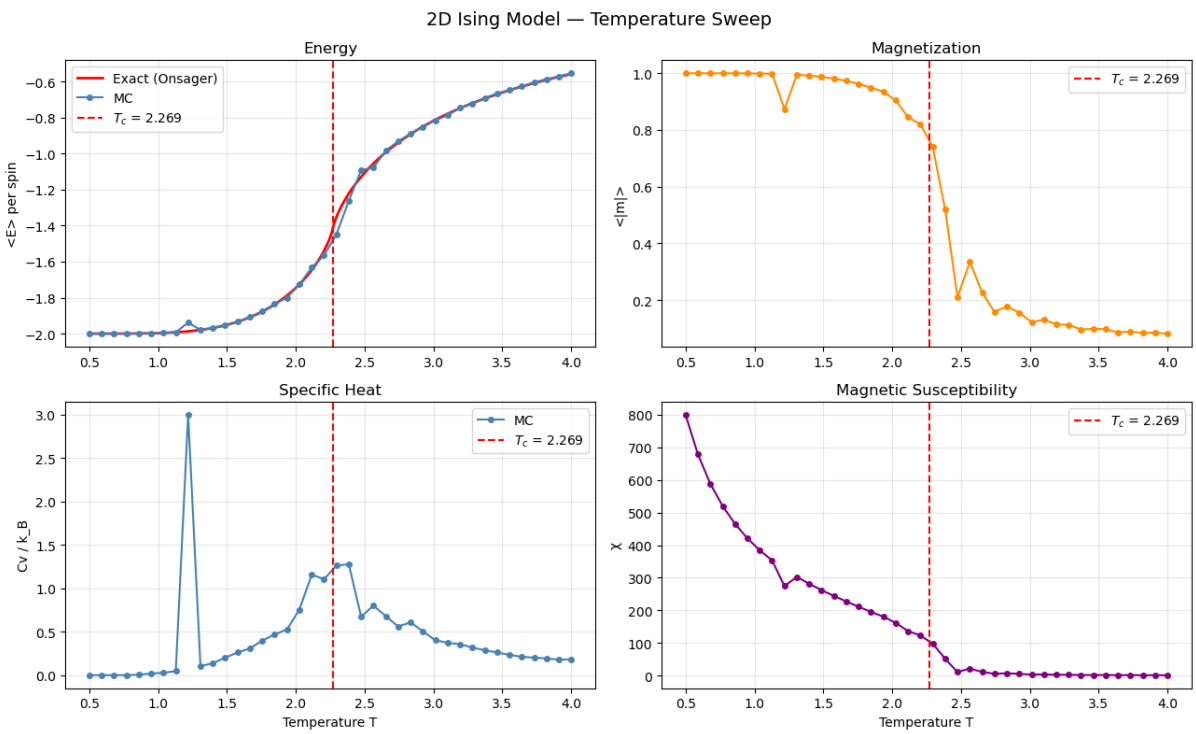


Figure 8: Thermodynamic observables for the 2D Ising model ($N = 20$). The red dashed line indicates the theoretical critical temperature $T_c \approx 2.269$.

16.1 Energy and the Onsager benchmark

The internal energy per spin $\langle e \rangle$ shows excellent agreement with the Onsager exact solution.

- At $T = 0.5$, the energy is fixed at -2.0, indicating a perfectly ordered ground state where every spin is aligned with its four neighbors.
- At the critical point $T_c = 2.269$, the simulation records an energy of approximately -1.444.

16.2 Magnetization and spontaneous symmetry breaking

The plot of $\langle |m| \rangle$ demonstrates the onset of spontaneous order.

- A sharp decline in magnetization is observed as the temperature passes T_c. By $T = 3.0$, the magnetization effectively collapses to near-zero values (≈ 0.1), characteristic of the paramagnetic phase.
- A prominent numerical artifact is visible around $T \approx 1.2$, where the magnetization suddenly drops. This occurs because, in a small 20×20 lattice, the system can switch between the all-up and all-down states during the measurement period, momentarily lowering the average absolute magnetization.

16.3 Response functions

The Specific Heat (C_v) and Susceptibility (χ) are the most sensitive indicators of the transition, as they are derived from fluctuations (variance).

- Specific heat: We observe a broad peak near $T_c \approx 2.269$ with a value of $C_v \approx 1.25$. This corresponds to the maximum energy fluctuations at the critical point.
- A massive, non-physical spike in C_v (reaching ≈ 3.0) appears at $T \approx 1.2$. This correlates exactly with the magnetization dip mentioned above; the global flipping of the lattice creates a massive variance in energy, resulting in a false peak.
- χ shows a dramatic divergence as $T \to 0$. In the ordered phase, the high susceptibility reflects the fact that the entire correlated lattice is poised to flip in unison, a behavior that would be suppressed in much larger systems or in the presence of an external field h.

16.4 Conclusion on Numerical Convergence

The agreement between our Monte Carlo energy data and the exact Onsager curve is a success. While finite-size effects (the artifacts at $T \approx 1.2$) are present due to the $N = 20$ lattice constraints, the fundamental physics of the second-order phase transition is clearly captured. To remove the false peaks and sharpen the transition at T_c, a larger lattice (e.g., $N = 64$) would be required, though at a significantly higher computational cost.

References

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[2] Metropolis, N., Rosenbluth, A. W., Rosenbluth, M. N., Teller, A. H., & Teller, E. (1953). Equation of state calculations by fast computing machines. *The Journal of Chemical Physics*, 21(6), 1087-1092.